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CS 621 DATA ANALYTICS

Machine Learning – Data Classification

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MACHINE LEARNING

- Machine learning is to learn patterns from data; we use training data to learn/train/construct a model.
- Then, we use the constructed/learned/trained model to predict new data.
- Three main areas:
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning

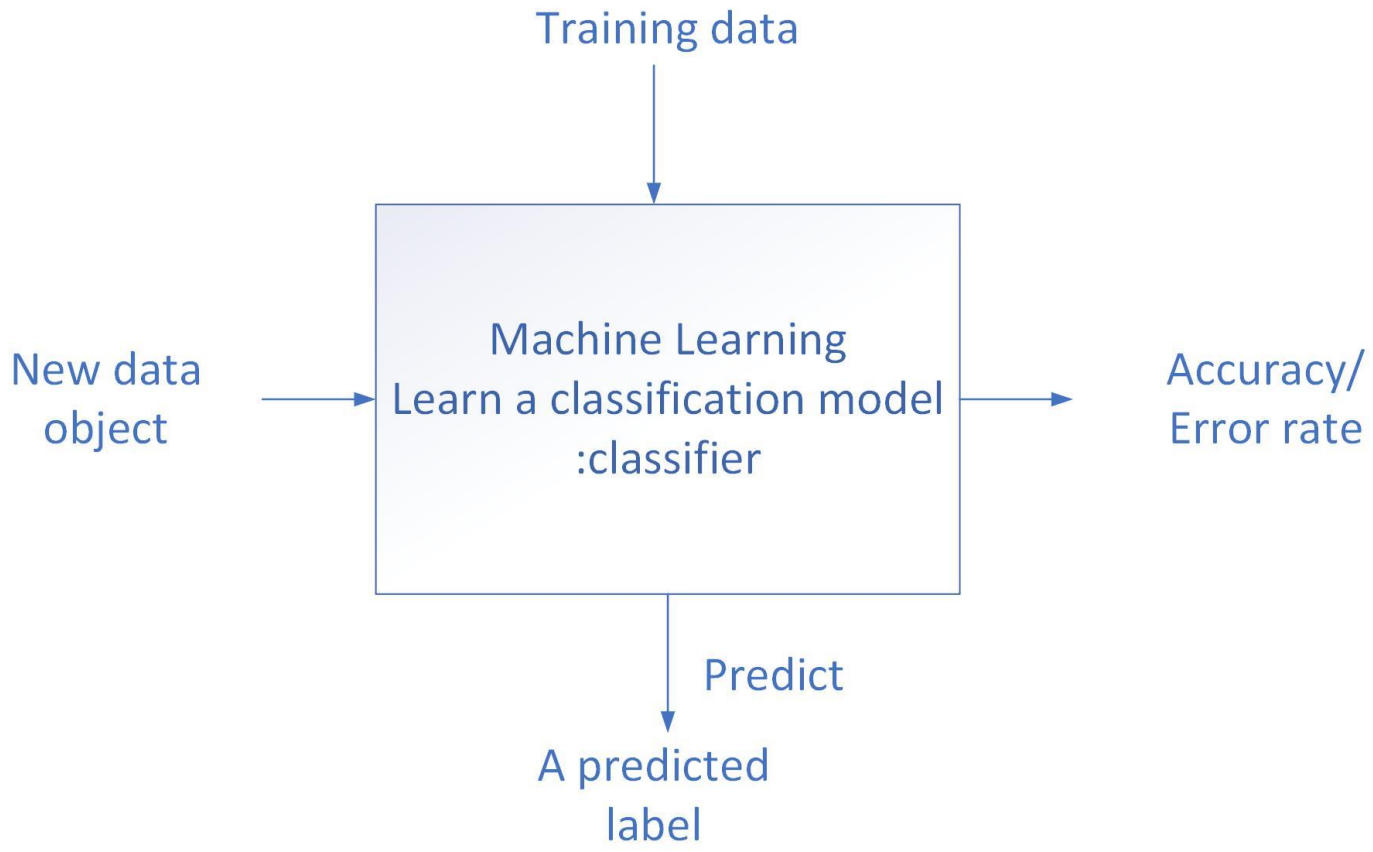
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DATA CLASSIFICATION

- Supervised Learning – data with class labels
- Training data provided with class labels.
- Goal is to predict class labels for new data based on the learned/constructed model.
- Examples:
  - To predict a team will win or lose in the new game.
  - To predict a newly received email is regular or spam.
  - To predict a customer's product review is positive or negative.

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CLASSIFICATION MODEL : THE CLASSIFIER



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BINARY OR MULTICLASS CLASSIFICATION

- Binary Classification: the number of labels to predict is two.
- Multiclass classification: to predict more than two labels. For example, we would like to predict tomorrow's weather as sunny, rainy, snowy, windy, etc.

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GENERAL CLASSIFICATION MODEL

	X1,	X2,	X3, ….,	Xm		Y
R1						C1
R2						C1
R3						C2
.						.
.						.
Rn						C1

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DIGIT CLASSIFICATION

- Assuming we work for the USPS, our job is to distribute mail by the zip code, e.g., 47306.
- We need to recognize digits in writing.



- And we have a lot of data available.
- Therefore, we would like to construct(learn) a digit recognition system so that it can recognize digits in the future.

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HANDWRITTEN DIGITS AS DATA

- We have many handwritten digits available; they are digital images; grayscale images;
- Say the image is 28x28, so
  - A pixel is stored in an integer list, size 784
  - The range of each integer is 0 to 255.
- We need to recognize any digit as 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9.

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HANDWRITTEN DIGIT IMAGES AS DATA

	X1,	X2,	X3, ….,	X784	Y(Label)
image1					7
image2					1
Image3					5
.					.
.					.
imagen					0

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IMAGE DATA IN CSV

label	pixel0	pixel1	pixel2	pixel3	pixel4	pixel5	pixel6	pixel7	pixel8	...	pixel774	pixel775	pixel776	pixel777	pixel778	pixel779	pixel780	pixel781	pixel782	pixel783	
0	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
3	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0		

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IMAGE, PANDAS, SCIKIT-LEARN, MATPLOTLIB

- Visually, a digit is an image.
- In Python, an image is an integer list or numpy array.
- We store each image as one row in csv (columns represents pixel positions).
- We load the csv file into Pandas.

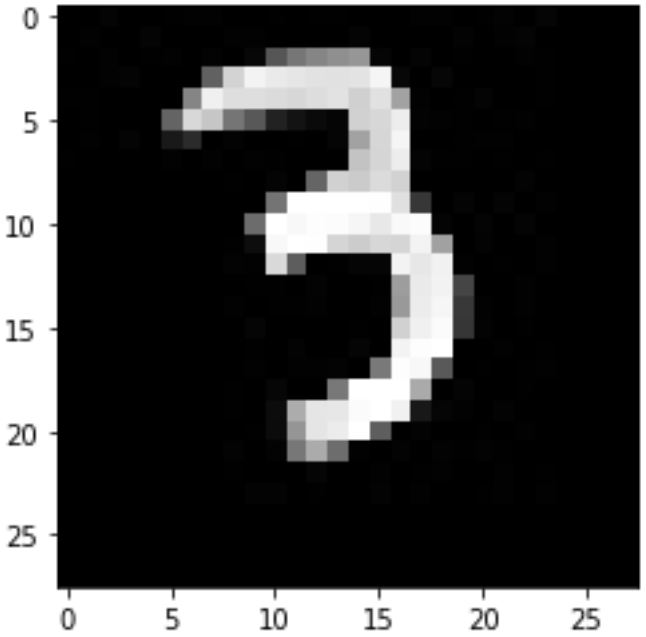
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IMAGE, PANDAS, SCIKIT-LEARN, MATPLOTLIB

- We use the scikit-learn decision tree as our classification algorithm.
- We use the training data to train/construct/learn the classification system.
- We then use the system to classify/predict a new image as 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9.
- A good classification system will have a high accuracy, whereas a not-so-great system may make lots of mistake.

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WRITTEN DIGIT THREE AS INTEGERS




```
0,0,2,2,0,0,0,0,4,0,0,0,0,3,0,2,4,0,0,0,0,0,1,0,2,0,1,2,0,3,0
50,5,0,4,0,1,0,0,2,1,0,0,0,0,0,0,2,4,0,0,0,98,211,243,234,230
0,6,0,0,0,0,1,0,0,0,0,1,1,0,0,0,0,133,240,222,222,219,215,221
0,1,4,0,0,1,0,0,0,0,0,1,0,0,0,100,216,198,117,88,34,19,10,5,2
1,2,0,0,0,0,0,0,2,0,4,0,25,43,0,3,0,0,1,0,5,162,214,245,0,0,1
0,0,0,0,2,0,2,0,2,1,0,0,0,195,213,237,4,1,1,0,1,0,1,0,0,0,0,0
4,0,101,209,203,219,210,0,0,0,0,0,3,0,0,0,0,0,0,0,0,0,0,0,0,0
5,255,254,224,55,0,4,0,4,0,2,0,0,0,0,0,0,0,0,0,0,0,0,0,108,25
32,249,252,4,0,4,0,4,0,0,0,0,0,0,0,0,0,0,0,0,0,1,13,248,255,2
244,161,0,3,0,0,2,0,0,0,0,0,0,0,0,0,0,0,0,3,2,217,93,0,1,5,5,
0,2,0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,3,0,0,1,157,234,242,69,0,
0,0,0,0,0,0,2,0,1,1,2,0,0,2,153,236,246,57,2,0,2,0,0,0,0,0,0,
0,0,0,0,1,0,209,240,251,54,3,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
53,255,0,0,2,0,1,0,0,0,0,0,0,0,0,0,0,0,0,2,0,2,2,3,0,122,25
0,0,0,0,0,0,0,0,0,0,0,0,1,0,6,0,0,120,255,255,255,170,0,4,0,0
0,0,0,0,0,1,0,12,172,229,231,251,255,241,24,5,2,0,3,0,0,0,0,0
0,1,11,149,229,236,255,94,0,3,0,0,1,0,2,0,0,0,0,0,0,0,0,0,0,0
109,2,0,3,0,3,1,3,1,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,5,0,0
```

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DEMONSTRATION

- >>>import numpy as np
- >>>from sklearn import tree
- >>>import matplotlib.pyplot as plot
- >>>import pandas as pd
- >>> …
- >>>clf = tree.DecisionTreeClassifier()
- >>>clf.fit(xtrain_all, train_all_label)
- >>>clf.predict(xtrain[8].reshape(1, -1))

Please follow the class code demonstration for details.



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